Set 23: Employee Performance Analysis

AIM

To analyze employee performance using R programming by creating a bar chart of employee distribution across departments, a line chart of performance scores, and a scatter plot to examine the relationship between years of service and performance scores.

ALGORITHM

1. Create a dataset containing Employee ID, Department, Years of Service, and Performance Score.
2. Load the ggplot2 library.
3. Create a bar chart showing the number of employees in each department.
4. Create a line chart showing employee performance scores.
5. Create a scatter plot to examine the relationship between years of service and performance scores.
6. Display the charts and interpret the results.

R PROGRAM

```
library(ggplot2)


employee <- data.frame(
  Employee_ID = 1:5,
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

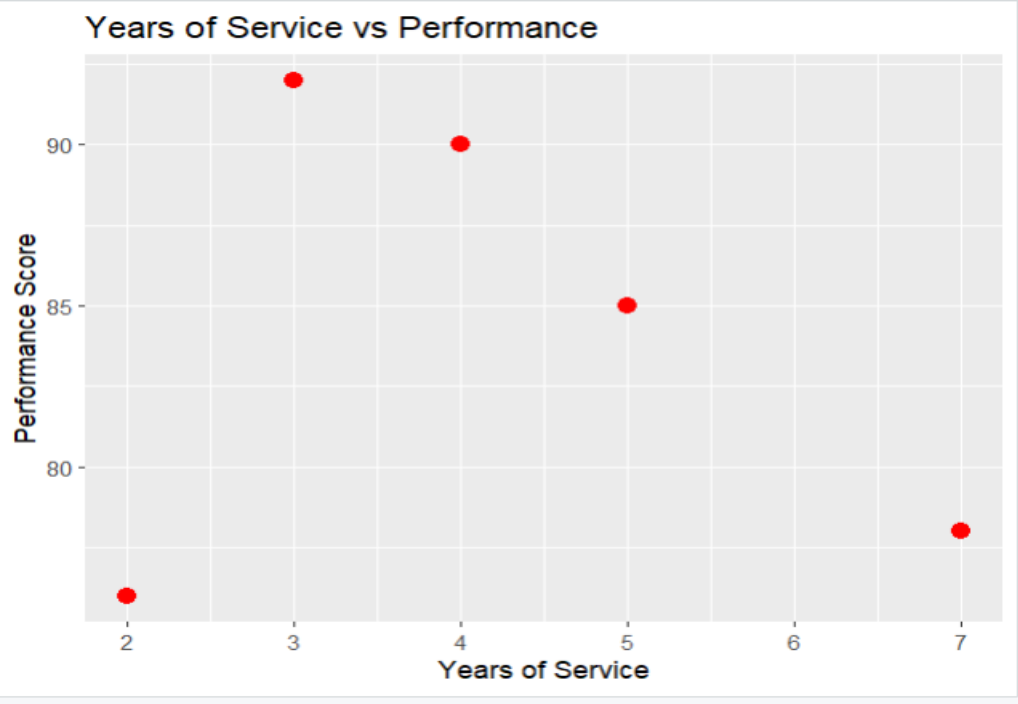

ggplot(employee, aes(Department, fill = Department)) +
  geom_bar() +
  labs(title = "Employees by Department",
       x = "Department", y = "Number of Employees")


ggplot(employee, aes(Employee_ID, Performance_Score, group = 1)) +
  geom_line(color = "blue") +
  geom_point(size = 3) +
```

```
  labs(title = "Employee Performance Trend",
       x = "Employee ID", y = "Performance Score")


ggplot(employee, aes(Years_Service, Performance_Score)) +
  geom_point(color = "red", size = 3) +
  labs(title = "Years of Service vs Performance",
       x = "Years of Service", y = "Performance Score")
```

OUTPUT:



RESULT

The bar chart shows the distribution of employees across departments. Sales and HR each have two employees, while Marketing has one. The line chart displays employee performance scores, and the scatter plot explores the relationship between years of service and performance. The highest performance score is 92, and the lowest is 76.